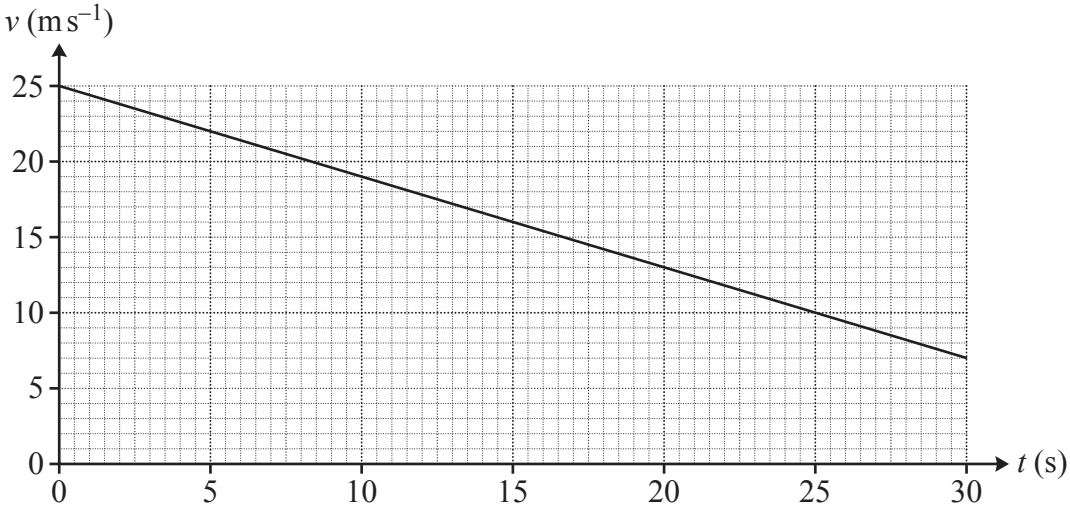
- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

9 Two trains are travelling in the same direction on parallel straight tracks and train A overtakes train B. At time t seconds after the front of train A overtakes the front of train B the velocities of trains A and B are $v_A\,\mathrm{m\,s^{-1}}$ and $v_B\,\mathrm{m\,s^{-1}}$ respectively.

The velocity of train A is modelled by $v_A = 25 - 0.6t$. The velocity-time graph of train A is shown below.



(a) A student argues that the speed of train A changes by $18\,\mathrm{m\,s^{-1}}$ in 30 seconds so its acceleration is $0.6\,\mathrm{m\,s^{-2}}$.

Comment on the validity of the student's argument. **[1]**

(b) When the front of train A overtakes the front of train B, train B has a velocity of $10\,\mathrm{m\,s^{-1}}$. The acceleration of train B is constant and is modelled as $0.15\,\mathrm{m\,s^{-2}}$.

Write down the equation for v_B in terms of t that models the velocity of train B. **[1]**

(c) Draw the velocity-time graph of train B on the copy of the diagram in the Printed Answer Booklet. **[1]**

(d) Determine the distance between the fronts of the trains at the time when the trains are travelling at the same velocity. **[3]**

(e) Explain why the model for train A would not be valid for large values of t. **[1]**